

Disease: Idiopathic pulmonary fibrosis (IPF)

Biological question: Which genes and pathways are differentially expressed in fibrotic IPF lung tissue compared with non-diseased donor lung tissue?

I would design a reproducible bulk RNA-seq workflow that begins with paired-end FASTQ files and ends with differential expression and pathway interpretation for the primary contrast IPF versus Control. After verifying sample metadata, I would run FastQC on raw reads and aggregate reports with MultiQC to screen for low base quality, adapter contamination, abnormal GC profiles, and outlier libraries. I would trim reads with fastp using paired-end adapter detection, a qualified quality threshold of Q20, a minimum retained length of 35 nucleotides, and polyG/polyX trimming because NovaSeq data commonly show terminal low-complexity artifacts. I would keep trimming conservative so informative sequence is not discarded unnecessarily.

For read assignment, I would use Salmon selective alignment rather than full genome alignment. Specifically, I would quantify against a decoy-aware transcriptome index built from the GRCh38 primary assembly and a matching GENCODE comprehensive transcript annotation. For stranded TruSeq mRNA libraries I would specify library type ISR instead of automatic inference, and I would enable `validateMappings` plus `sequence`, `GC`, and `positional bias` correction. I chose Salmon because it is fast, handles multi-mapped reads probabilistically, and performs well for gene-level differential expression when transcript estimates are summarized with `tximport`. I would import Salmon quantifications into R with `tximport` using `countsFromAbundance = "lengthScaledTPM"` to reduce transcript-length bias during transcript-to-gene summarization. Reference files and databases would include GRCh38 and GENCODE for the transcriptome and annotation, plus MSigDB Hallmark and Reactome gene sets for GSEA and GO Biological Process, Reactome, and KEGG through `gprofiler2` for over-representation analysis.

For differential expression, I would analyze gene counts in DESeq2 with the design formula $\sim \text{batch} + \text{sex} + \text{condition}$, where condition is IPF or Control. This design adjusts for known technical batch effects and a major biological covariate without overcomplicating the model. Before fitting, I would prefilter genes by retaining those with at least 10 counts in at least 25% of samples to reduce noise from minimally expressed features. I would test the main IPF-versus-Control contrast with DESeq2's Wald framework, control false discovery with Benjamini-Hochberg FDR at 0.05, and apply `apeglm` shrinkage to log2 fold changes so effect sizes are more stable and interpretable.

For pathway analysis, I would run `fgseaMultilevel` on a ranked list based on the DESeq2 Wald statistic because it preserves both direction and strength of differential signal without imposing an arbitrary significance cutoff. I would use `minSize = 15` and `maxSize = 500` to avoid unstable very small sets and overly broad, non-specific pathways. I would complement GSEA with `gprofiler2` over-representation analysis on significant genes using the expressed-gene universe as the

custom background, rather than the whole genome, to better match the tested feature space. Nondefault settings worth justifying are the expressed-gene background in gprofiler2, bias-correction flags in Salmon, explicit strandedness, and the gene prefilter used before DESeq2.

Biologically, the IPF-versus-Control contrast addresses which transcripts mark extracellular matrix deposition, TGF-beta signaling, epithelial stress, immune activation, and altered repair programs in fibrotic lung. If metadata allow, secondary contrasts could compare rapid versus slow progressors or adjust for smoking history and RNA integrity. Important limitations include bulk-tissue heterogeneity, differences in cell composition between diseased and donor lungs, possible confounding by smoking or ischemic time, RNA degradation, annotation-dependent quantification, and the fact that gene-level summarization may miss isoform-specific biology. Pathway databases are also incomplete and biased toward well-studied genes, so enrichment should guide interpretation rather than serve as proof of mechanism.